

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and deployed **production-grade AI/LLM-powered solutions** using **Python**, **FastAPI**, and **LangChain** orchestration frameworks to automate transaction-flow intelligence and operational alerting across enterprise banking systems, reducing manual intervention by **40%** and accelerating incident resolution by **35%**.
- Built and optimized **high-performance backend systems and REST APIs** in **Java Spring Boot** and **Python**, integrating **LLM inference engines** with existing SWIFT messaging platforms and ERP business systems, achieving **99.8% uptime** and sub-500ms latency for real-time payment-operations analytics.
- Architected and optimized **data pipelines** using **Python**, **pandas**, **NumPy**, and **scikit-learn** to ingest, transform, and train machine learning models on **petabyte-scale** financial transaction datasets, enabling AI systems to power intelligent applications across cross-border payments, SEPA, and ISO 20022 message-flow prediction with **92% accuracy**.
- Deployed, managed, and scaled **AI/LLM systems within AWS cloud infrastructure (EC2, Lambda, CloudWatch)** and implemented **MLOps practices** for continuous model monitoring, evaluation, and optimization in production, reducing inference latency by **28%** and cutting infrastructure costs by **22%** through resource auto-scaling.
- Monitored, evaluated, and continuously improved **system performance in production** using **CloudWatch**, **Dynatrace**, and **Splunk** dashboards; implemented alerting logic for AI model drift detection and SLA compliance, maintaining **99.9% service availability** and enabling real-time performance tuning across distributed systems.
- Collaborated across engineering, product, and domain teams to integrate **AI seamlessly into existing platforms**; conducted UAT, change-management coordination, and post-deployment triage; documented system architecture and MLOps workflows to ensure reproducibility and knowledge transfer across **50+ stakeholders** in regulated banking environments.

SKILLS

AI & Machine Learning – LLM architecture, LangChain, prompt engineering, model deployment, scikit-learn, supervised/unsupervised learning, model evaluation metrics, AI system design.

Backend Engineering & APIs – Python, Java Spring Boot, FastAPI, Flask, REST API design, system architecture, microservices, distributed systems.

Data Engineering & Pipelines – Data pipeline architecture, ETL/ELT, pandas, NumPy, data warehousing, data modelling, SQL (PostgreSQL, MySQL, Oracle), real-time data ingestion.

Cloud Infrastructure & Deployment – AWS (EC2, S3, Lambda, CloudWatch), system deployment, scaling, auto-scaling, cloud architecture, distributed computing.

MLOps & Monitoring – Model monitoring, performance optimization, system observability, Splunk, Dynatrace, CloudWatch, alerting, SLA management, production reliability.

Engineering Practices – CI/CD (Jenkins, Git), code review, change management, Docker, Linux, Jira, Agile/Scrum.

Domain Expertise – Banking & financial services, SWIFT messaging (MT/MX), ISO 20022, SEPA, cross-border payments, PSD2, Open Banking, ERP integration, payments operations, regulatory compliance (GDPR).

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Intelligent Payment Operations Platform – AI-Powered Transaction Intelligence Engine (HCL Technologies)

- Designed and deployed an **end-to-end AI/LLM solution** that ingested real-time **SWIFT MT/MX message streams** and **ISO 20022 payment flows**, using **LangChain** orchestration and **fine-tuned LLM models** to predict transaction anomalies, classify payment types, and auto-generate operational alerts for **500K+ daily payments** across cross-border and SEPA corridors.
- Built **high-performance backend APIs in Python FastAPI and Java Spring Boot** to serve **LLM inference** at scale; architected **distributed data pipelines using pandas, NumPy, and scikit-learn** to featurize payment metadata, train ensemble models on historical transaction data, and achieve **94% fraud-detection accuracy** with **<200ms inference latency**.
- Deployed the platform on **AWS infrastructure (EC2, Lambda, S3, CloudWatch)** with auto-scaling policies; implemented **MLOps monitoring via Splunk and Dynatrace** to track model drift, system health, and SLA compliance; reduced manual incident triage by **45%** and decreased false-positive alerts by **31%** through continuous model retraining.
- Integrated AI system seamlessly with legacy **ERP business systems** and enterprise banking platforms; conducted **change-management coordination** across operations, compliance, and engineering teams; documented system architecture, API contracts, and deployment runbooks to enable **6-month operational handoff** to support teams in a regulated environment.

Real-Time Financial Analytics Dashboard – Data Pipeline to Production Reporting (HCL Technologies)

- Architected and optimized **end-to-end data pipelines** using **Python, pandas, PostgreSQL, and Oracle** to extract, transform, and load multi-terabyte datasets from SWIFT messaging systems and payment-operations databases into a **centralized data warehouse**, processing **2M+ transactions daily** with **sub-5-minute latency**.
- Built **production-grade monitoring dashboards and analytical reports** using **Power BI and DAX** to surface real-time KPIs (transaction throughput, payment latency, SLA compliance, regulatory metrics) for banking operations and compliance teams; enabled self-service analytics for **40+ business stakeholders** and reduced manual reporting overhead by **60%**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)